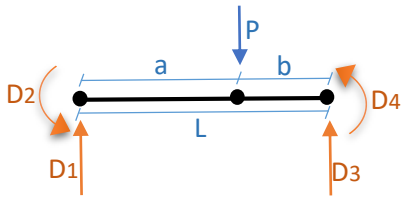
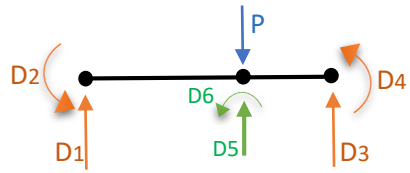
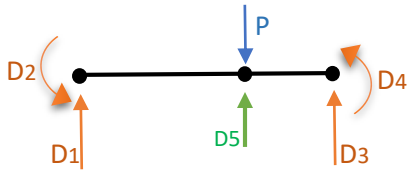
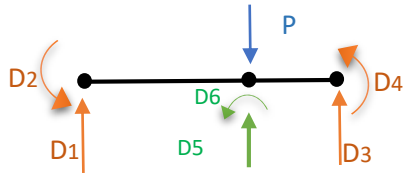
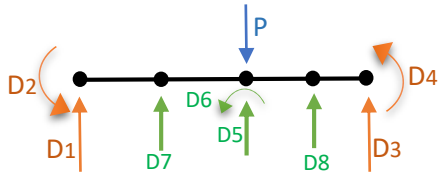
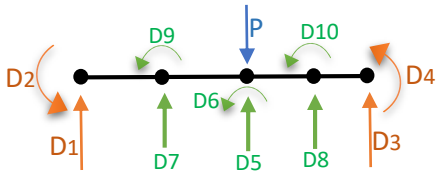
تمرین سری سوم (روش های المان محدود)

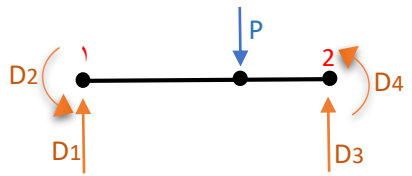
(دانشجو: سبا آزادگان / شماره دانشجویی: ۶۰۲[redacted]۹۷)

1. بر اساس ۴ مش داده شده و نیز یک یا دو مش انتخابی ریزتر، این مسئله را به روش اجزا محدود تحلیل نمایید.

	Mesh of 1 lower order E.B Beam F.E	مش های داده شده
	Mesh of 2 Lower order E.B Beam F.E	
	Mesh of 1 Higher order E.B Beam F.E	
	Mesh of 1 Higher order E.B Beam F.E	

	**مش اضافه شده** Mesh of 2 Higher order E.B Beam F.E
	**مش انتخابی** Mesh of 4 Lower order E.B Beam F.E

حل:

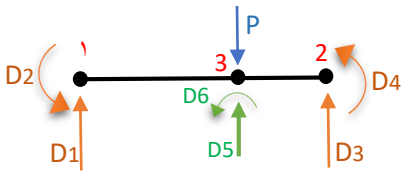


Mesh of
1 lower order E.B Beam F.E

$$[K]_{1-2} = \begin{bmatrix} \frac{12EI}{l^3} & \frac{6EI}{l^2} & -\frac{12EI}{l^3} & \frac{6EI}{l^2} \\ \frac{6EI}{l^2} & \frac{4EI}{l} & -\frac{6EI}{l^2} & \frac{2EI}{l} \\ -\frac{12EI}{l^3} & -\frac{6EI}{l^2} & \frac{12EI}{l^3} & -\frac{6EI}{l^2} \\ \frac{6EI}{l^2} & \frac{2EI}{l} & -\frac{6EI}{l^2} & \frac{4EI}{l} \end{bmatrix}$$

$$[K]_{structure} = \begin{bmatrix} \frac{12EI}{l^3} & \frac{6EI}{l^2} & -\frac{12EI}{l^3} & \frac{6EI}{l^2} \\ \frac{6EI}{l^2} & \frac{4EI}{l} & -\frac{6EI}{l^2} & \frac{2EI}{l} \\ -\frac{12EI}{l^3} & -\frac{6EI}{l^2} & \frac{12EI}{l^3} & -\frac{6EI}{l^2} \\ \frac{6EI}{l^2} & \frac{2EI}{l} & -\frac{6EI}{l^2} & \frac{4EI}{l} \end{bmatrix}$$

$$\{D\}_{E.l\ 1-2} = \begin{Bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{Bmatrix} \ and \ \begin{Bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{Bmatrix} = [K]_{1-2} \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$



Mesh of
2 Lower order E.B Beam F.E

$$[K]_{1-3} = \begin{array}{c} \begin{array}{cccc} D1 & D2 & D5 & D6 \end{array} \\ \begin{bmatrix} \frac{12EI}{a^3} & \frac{6EI}{a^2} & -\frac{12EI}{a^3} & \frac{6EI}{a^2} \\ \frac{6EI}{a^2} & \frac{4EI}{a} & -\frac{6EI}{a^2} & \frac{2EI}{a} \\ -\frac{12EI}{a^3} & -\frac{6EI}{a^2} & \frac{12EI}{a^3} & -\frac{6EI}{a^2} \\ \frac{6EI}{a^2} & \frac{2EI}{a} & -\frac{6EI}{a^2} & \frac{4EI}{a} \end{bmatrix} \end{array}$$

$$[K]_{3-2} = \begin{array}{c} \begin{array}{cccc} D5 & D6 & D3 & D4 \end{array} \\ \begin{bmatrix} \frac{12EI}{b^3} & \frac{6EI}{b^2} & -\frac{12EI}{b^3} & \frac{6EI}{b^2} \\ \frac{6EI}{b^2} & \frac{4EI}{b} & -\frac{6EI}{b^2} & \frac{2EI}{b} \\ -\frac{12EI}{b^3} & -\frac{6EI}{b^2} & \frac{12EI}{b^3} & -\frac{6EI}{b^2} \\ \frac{6EI}{b^2} & \frac{2EI}{b} & -\frac{6EI}{b^2} & \frac{4EI}{b} \end{bmatrix} \end{array}$$

$$[K]_{structure} = \begin{bmatrix} \frac{12EI}{a^3} & \frac{4EI}{a} & & & & \\ & & \frac{12EI}{b^3} & \frac{4EI}{b} & & \\ & & & & 12EI(\frac{1}{a^3}+\frac{1}{b^3}) & -6EI(\frac{1}{a^2}-\frac{1}{b^2}) \\ & & & & -6EI(\frac{1}{a^2}-\frac{1}{b^2}) & 4EI(\frac{1}{a}+\frac{1}{b}) \end{bmatrix}$$

$$D_5 : P_{D_5} = -P$$
$$D_6 : P_{D_6} = 0$$
$$[S]\{D\} = \{P\} \equiv [K]\{U\} = \{P\}$$

$$EI \begin{bmatrix} 12(\frac{1}{a^3}+\frac{1}{b^3}) & -6(\frac{1}{a^2}-\frac{1}{b^2}) \\ -6(\frac{1}{a^2}-\frac{1}{b^2}) & 4(\frac{1}{a}+\frac{1}{b}) \end{bmatrix} \begin{Bmatrix} D_5 \\ D_6 \end{Bmatrix} = \begin{Bmatrix} -P \\ 0 \end{Bmatrix}$$

$$D_5 = -\frac{a^3 b^3 p}{3EI(a + b)(a^2 + 2ab + b^2)}$$

$$D_6 = \frac{a^2 b^2 p(a - b)}{2EI(a^3 + 3a^2 b + 3ab^2 + b^3)}$$

$$\begin{Bmatrix} P_1 \\ P_2 \\ P_5 \\ P_6 \end{Bmatrix} = [K]_{1-3} \begin{Bmatrix} D_1 \\ D_2 \\ D_5 \\ D_6 \end{Bmatrix}$$

$$\begin{Bmatrix} P_1 \\ P_2 \\ P_5 \\ P_6 \end{Bmatrix} = \begin{bmatrix} \frac{12EI}{a^3} & \frac{6EI}{a^2} & -\frac{12EI}{a^3} & \frac{6EI}{a^2} \\ \frac{6EI}{a^2} & \frac{4EI}{a} & -\frac{6EI}{a^2} & \frac{2EI}{a} \\ -\frac{12EI}{a^3} & -\frac{6EI}{a^2} & \frac{12EI}{a^3} & -\frac{6EI}{a^2} \\ \frac{6EI}{a^2} & \frac{2EI}{a} & -\frac{6EI}{a^2} & \frac{4EI}{a} \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ -\frac{a^3b^3p}{3EI(a+b)(a^2+2ab+b^2)} \\ \frac{a^2b^2p(a-b)}{2EI(a^3+3a^2b+3ab^2+b^3)} \end{Bmatrix}$$

$$P_1 = \frac{pb^2}{(a+b)^3}(3a+b)$$

$$P_2 = \frac{pab^2}{(a+b)^2}$$

$$P_5 = -\frac{pb^2}{(a+b)^3}(3a+b)$$

$$P_6 = \frac{2pa^2b^2}{(a+b)^3}$$

$$\begin{Bmatrix} P_5 \\ P_6 \\ P_3 \\ P_4 \end{Bmatrix} = [K]_{3-2} \begin{Bmatrix} D_5 \\ D_6 \\ D_3 \\ D_4 \end{Bmatrix}$$

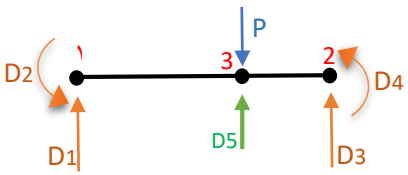
$$\begin{Bmatrix} P_5 \\ P_6 \\ P_3 \\ P_4 \end{Bmatrix} = [K]_{3-2} = \begin{bmatrix} \frac{12EI}{b^3} & \frac{6EI}{b^2} & -\frac{12EI}{b^3} & \frac{6EI}{b^2} \\ \frac{6EI}{b^2} & \frac{4EI}{b} & -\frac{6EI}{b^2} & \frac{2EI}{b} \\ -\frac{12EI}{b^3} & -\frac{6EI}{b^2} & \frac{12EI}{b^3} & -\frac{6EI}{b^2} \\ \frac{6EI}{b^2} & \frac{2EI}{b} & -\frac{6EI}{b^2} & \frac{4EI}{b} \end{bmatrix} \begin{Bmatrix} -\frac{a^3b^3p}{3EI(a+b)(a^2+2ab+b^2)} \\ \frac{a^2b^2p(a-b)}{2EI(a^3+3a^2b+3ab^2+b^3)} \\ 0 \\ 0 \end{Bmatrix}$$

$$P_6 = -\frac{2pa^2b^2}{(a+b)^3}$$

$$P_3 = \frac{pa^2}{(a+b)^3}(a+3b)$$

$$P_4 = -\frac{pa^2b}{(a+b)^2}$$

$$P_5 = -\frac{pa^2}{(a+b)^3}(a+3b)$$



Mesh of
1 Higher order E.B Beam F.E

$$D_1 = \varphi_{(x=0)} = \alpha_1$$
$$D_2 = \varphi'_{(x=0)} = \alpha_2$$
$$D_3 = \varphi_{(x=l)} = \alpha_1 + \alpha_2 l + \alpha_3 l^2 + \alpha_4 l^3 + \alpha_5 l^5$$
$$D_4 = \varphi'_{(x=l)} = \alpha_2 + 2\alpha_3 l + 3\alpha_4 l^2 + 5\alpha_5 l^4$$
$$D_5 = \varphi_{(x=a)} = \alpha_1 + \alpha_2 (a) + \alpha_3 (a)^2 + \alpha_4 (a)^3 + \alpha_5 (a)^5$$

$$\boldsymbol{\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^5}$$

$$\varphi'_{(x)} = 0 + \alpha_2 + 2\alpha_3 x + 3\alpha_4 x^2 + 5\alpha_5 x^4$$

$$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & l & l^2 & l^3 & l^5 \\ 0 & 1 & 2l & 3l^2 & 5l^4 \\ 1 & a & a^2 & a^3 & a^5 \end{bmatrix}$$

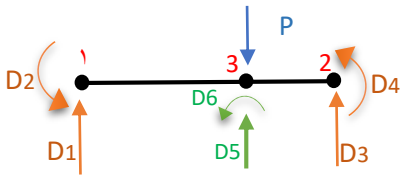
$$[A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -\dfrac{3a^3 + 6a^2 l + 4al^2 + 2l^3}{a^3 l^2 + 2a^2 l^3} & -2\dfrac{a^2 + 2al + l^2}{a^2 l + 2al^2} & -\dfrac{-3a^3 + 5al^2}{a^3 l^2 - 3al^4 + 2l^5} & -\dfrac{a^2 + al}{a^2 l + al^2 - 2l^3} & \dfrac{2l^3}{a^5 - 3a^3 l^2 + 2a^2 l^3} \\ \dfrac{2a^3 + 4a^2 l + 6al^2 + 3l^3}{a^3 l^3 + 2a^2 l^4} & \dfrac{a^2 + 2al + 3l^2}{a^2 l^2 + 2al^3} & -\dfrac{2a^3 - 5l^3}{a^3 l^3 - 3al^5 + 2l^6} & \dfrac{a^2 + al + l^2}{a^2 l^2 + al^3 - 2l^4} & -\dfrac{3l^2}{a^5 - 3a^3 l^2 + 2a^2 l^3} \\ -\dfrac{2a + l}{a^3 l^3 + 2a^2 l^4} & -\dfrac{1}{a^2 l^2 + 2al^3} & \dfrac{2a - 3l}{a^3 l^3 - 3al^5 + 2l^6} & -\dfrac{1}{a^2 l^2 + al^3 - 2l^4} & \dfrac{1}{a^5 - 3a^3 l^2 + 2a^2 l^3} \end{bmatrix}$$

$$[P] = [1 \quad x \quad x^2 \quad x^3 \quad x^5]$$

$$[N_{(x)}] = [P][A_E]^{-1}$$

$$N''_{1(x)} = -\frac{3a^3 + 6a^2 l + 4al^2 + 2l^3}{a^3 l^2 + 2a^2 l^3}(2) + \frac{2a^3 + 4a^2 l + 6al^2 + 3l^3}{a^3 l^3 + 2a^2 l^4}(6x) - \frac{2a + l}{a^3 l^3 + 2a^2 l^4}(20x^3)$$
$$N''_{2(x)} = -2\frac{a^2 + 2al + l^2}{a^2 l + 2al^2}(2) + \frac{a^2 + 2al + 3l^2}{a^2 l^2 + 2al^3}(6x) - \frac{1}{a^2 l^2 + 2al^3}(20x^3)$$
$$N''_{3(x)} = -\frac{-3a^3 + 5al^2}{a^3 l^2 - 3al^4 + 2l^5}(2) - \frac{2a^3 - 5l^3}{a^3 l^3 - 3al^5 + 2l^6}(6x) + \frac{2a - 3l}{a^3 l^3 - 3al^5 + 2l^6}(20x^3)$$
$$N''_{4(x)} = -\frac{a^2 + al}{a^2 l + al^2 - 2l^3}(2) + \frac{a^2 + al + l^2}{a^2 l^2 + al^3 - 2l^4}(6x) - \frac{1}{a^2 l^2 + al^3 - 2l^4}(20x^3)$$
$$N''_{5(x)} = \frac{2l^3}{a^5 - 3a^3 l^2 + 2a^2 l^3}(2) - \frac{3l^2}{a^5 - 3a^3 l^2 + 2a^2 l^3}(6x) + \frac{1}{a^5 - 3a^3 l^2 + 2a^2 l^3}(20x^3)$$

$$[K]_F = EI \int_0^l [N''(x)]^T [N''(x)] dx \rightarrow$$ این انتگرال دارای جواب نمی باشد .



Mesh of
1 Higher order E.B Beam F.E

$$D_1 = \varphi_{(x=0)} = \alpha_1$$
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$$D_3 = \varphi_{(x=l)} = \alpha_1 + \alpha_2 l + \alpha_3 l^2 + \alpha_4 l^3 + \alpha_5 l^4 + \alpha_6 l^5$$
$$D_4 = \varphi'_{(x=l)} = \alpha_2 + 2\alpha_3 l + 3\alpha_4 l^2 + 4\alpha_5 l^3 + 5\alpha_6 l^4$$
$$D_5 = \varphi_{(x=a)} = \alpha_1 + \alpha_2(a) + \alpha_3(a)^2 + \alpha_4(a)^3 + \alpha_5(a)^4 + +\alpha_6(a)^5$$
$$D_6 = \varphi'_{(x=a)} = \alpha_2 + 2\alpha_3(a) + 3\alpha_4(a)^2 + 4\alpha_5(a)^3 + 5\alpha_6(a)^4$$

$$\boldsymbol{\varphi(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^4 + \alpha_6 x^5}$$

$$\varphi'_{(x)} = 0 + \alpha_2 + 2\alpha_3 x + 3\alpha_4 x^2 + 4\alpha_5 x^3 + 5\alpha_6 x^4$$

$$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & l & l^2 & l^3 & l^4 & l^5 \\ 0 & 1 & 2l & 3l^2 & 4l^3 & 5l^4 \\ 1 & a & a^2 & a^3 & a^4 & a^5 \\ 0 & 1 & 2a & 3a^2 & 4a^3 & 5a^4 \end{bmatrix}$$

$$[A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{3a^2 + 4al + 3l^2}{a^2 l^2} & -\frac{2(a + l)}{al} & -\frac{-3a^3 + 5la^2}{a^3 l^2 - 3a^2 l^3 + 3al^4 - l^5} & -\frac{a^2}{a^2 l - 2al^2 + l^3} & -\frac{-3l^3 + 5al^2}{-a^5 + 3a^4 l - 3a^3 l^2 + a^2 l^3} & -\frac{l^2}{a^3 - 2a^2 l + al^2} \\ \frac{2(a^3 + 4a^2 l + 4al^2 + l^3)}{a^3 l^3} & \frac{a^2 + 4al + l^2}{a^2 l^2} & -\frac{2(a^3 + a^2 l - 5al^2)}{a^3 l^3 - 3a^2 l^4 + 3al^5 - l^6} & \frac{a^2 + 2la}{a^2 l^2 - 2al^3 + l^4} & -\frac{2(-5a^2 l + al^2 + l^3)}{-a^6 + 3a^5 l - 3a^4 l^2 + a^3 l^3} & \frac{l^2 + 2al}{a^4 - 2a^3 l + a^2 l^2} \\ -\frac{4a^2 + 7al + 4l^2}{a^3 l^3} & -\frac{2(a + l)}{a^2 l^2} & -\frac{-4a^2 + 5al + 5l^2}{a^3 l^3 - 3a^2 l^4 + 3al^5 - l^6} & -\frac{2a + l}{a^2 l^2 - 2al^3 + l^4} & -\frac{5a^2 + 5al - 4l^2}{-a^6 + 3a^5 l - 3a^4 l^2 + a^3 l^3} & -\frac{a + 2l}{a^4 - 2a^3 l + a^2 l^2} \\ \frac{2(a + l)}{a^3 l^3} & \frac{1}{a^2 l^2} & -\frac{2(a - 2l)}{a^3 l^3 - 3a^2 l^4 + 3al^5 - l^6} & \frac{1}{a^2 l^2 - 2al^3 + l^4} & \frac{2(2a - l)}{-a^6 + 3a^5 l - 3a^4 l^2 + a^3 l^3} & \frac{1}{a^4 - 2a^3 l + a^2 l^2} \end{bmatrix}$$

$$[P] = [1 \quad x \quad x^2 \quad x^3 \quad x^4 \quad x^5]$$

$$[N_{(x)}] = [P][A_E]^{-1}$$

$$N''_{1(x)} = -\frac{3a^2 + 4al + 3l^2}{a^2l^2}(2) + \frac{2(a^3 + 4a^2l + 4al^2 + l^3)}{a^3l^3}(6x) - \frac{4a^2 + 7al + 4l^2}{a^3l^3}(12x^2) + \frac{2(a + l)}{a^3l^3}(20x^3)$$

$$N''_{2(x)} = -\frac{2(a + l)}{al}(2) + \frac{a^2 + 4al + l^2}{a^2l^2}(6x) - \frac{2(a + l)}{a^2l^2}(12x^2) + \frac{1}{a^2l^2}(20x^3)$$

$$N''_{3(x)} = -\frac{-3a^3 + 5la^2}{a^3l^2 - 3a^2l^3 + 3al^4 - l^5}(2) - \frac{2(a^3 + a^2l - 5al^2)}{a^3l^3 - 3a^2l^4 + 3al^5 - l^6}(6x) - \frac{-4a^2 + 5al + 5l^2}{a^3l^3 - 3a^2l^4 + 3al^5 - l^6}(12x^2) - \frac{2(a - 2l)}{a^3l^3 - 3a^2l^4 + 3al^5 - l^6}(20x^3)$$

$$N''_{4(x)} = -\frac{a^2}{a^2l - 2al^2 + l^3}(2) + \frac{a^2 + 2la}{a^2l^2 - 2al^3 + l^4}(6x) - \frac{2a + l}{a^2l^2 - 2al^3 + l^4}(12x^2) + \frac{1}{a^2l^2 - 2al^3 + l^4}(20x^3)$$

$$N''_{5(x)} = -\frac{-3l^3 + 5al^2}{-a^5 + 3a^4l - 3a^3l^2 + a^2l^3}(2) - \frac{2(-5a^2l + al^2 + l^3)}{-a^6 + 3a^5l - 3a^4l^2 + a^3l^3}(6x) - \frac{5a^2 + 5al - 4l^2}{-a^6 + 3a^5l - 3a^4l^2 + a^3l^3}(12x^2) + \frac{2(2a - l)}{-a^6 + 3a^5l - 3a^4l^2 + a^3l^3}(20x^3)$$

$$N''_{6(x)} = -\frac{l^2}{a^3 - 2a^2l + al^2}(2) + \frac{l^2 + 2al}{a^4 - 2a^3l + a^2l^2}(6x) - \frac{a + 2l}{a^4 - 2a^3l + a^2l^2}(12x^2) + \frac{1}{a^4 - 2a^3l + a^2l^2}(20x^3)$$

$$[K]_F = EI\int_0^l [N''(x)]^T [N''(x)]dx$$

$$[K]_F = EI\begin{bmatrix} & \\ \boxed{\begin{matrix} \frac{4l^5(175a^4 - 350a^3l + 265a^2l^2 - 90al^3 + 12l^4)}{35a^6(a - l)^6} & -\frac{2l^5(70a^3 - 105a^2l + 59al^2 - 12l^3)}{35a^5(a - l)^5} \\ -\frac{2l^5(70a^3 - 105a^2l + 59al^2 - 12l^3)}{35a^5(a - l)^5} & \frac{4l^5(147a^4 - 595a^3l + 759a^2l^2 - 320al^3 + 47l^4)}{35a^4(a - l)^4(a - 2l)^2} \end{matrix}} \end{bmatrix}$$

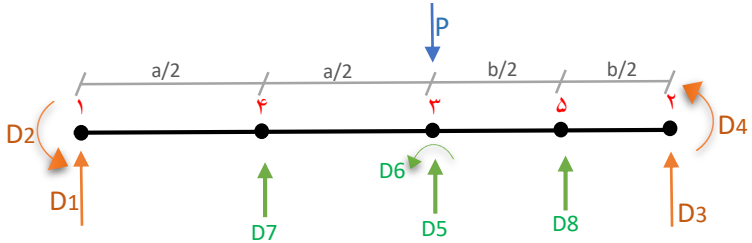
$$D_5: P_{D_5} = -P$$
$$D_6: P_{D_6} = 0$$
$$[S]\{D\} = \{P\} \equiv [K]\{U\} = \{P\}$$

$$EI\begin{bmatrix} \frac{4l^5(175a^4 - 350a^3l + 265a^2l^2 - 90al^3 + 12l^4)}{35a^6(a - l)^6} & -\frac{2l^5(70a^3 - 105a^2l + 59al^2 - 12l^3)}{35a^5(a - l)^5} \\ -\frac{2l^5(70a^3 - 105a^2l + 59al^2 - 12l^3)}{35a^5(a - l)^5} & \frac{4l^5(147a^4 - 595a^3l + 759a^2l^2 - 320al^3 + 47l^4)}{35a^4(a - l)^4(a - 2l)^2} \end{bmatrix}\begin{Bmatrix} D_5 \\ D_6 \end{Bmatrix} = \begin{Bmatrix} -P \\ 0 \end{Bmatrix}$$

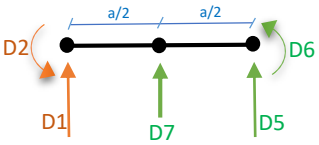
$$D5 = \frac{35a^6p(a - l)^6(175a^4 - 350a^3l + 265a^2l^2 - 90al^3 + 12l^4)}{EIl^2(4900a^{10}l^6 - 24500a^9l^7 + 53585a^8l^8 - 122500a^8 - 67340a^7l^9 + 490000a^7l + 53426a^6l^{10} - 861000a^6l^2 - 27488a^5l^{11} + 868000a^5l^3 + 8977a^4l^{12} - 549700a^4l^4 - 1704a^3l^{13} + 224400a^3l^5 + 144a^2l^{14} - 57840a^2l^6 + 8640al^7 - 576l^8)}$$

$$D6 = \frac{35a^7lp(a - l)^7(70a^3 - 105a^2l + 59al^2 - 12l^3)}{2EI(4900a^{10}l^6 - 24500a^9l^7 + 53585a^8l^8 - 122500a^8 - 67340a^7l^9 + 490000a^7l + 53426a^6l^{10} - 861000a^6l^2 - 27488a^5l^{11} + 868000a^5l^3 + 8977a^4l^{12} - 549700a^4l^4 - 1704a^3l^{13} + 224400a^3l^5 + 144a^2l^{14} - 57840a^2l^6 + 8640al^7 - 576l^8)}$$

$$P_x = (-P)N_{(x)} \rightarrow P_1, P_2, P_3, P_4, P_5, P_6 \ \surd\surd$$



Mesh of
2 Higher order E.B Beam F.E



E.L 1-3

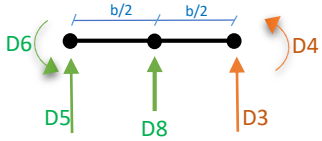
$$D_1 = \varphi_{(x=0)} = \alpha_1$$

$$D_2 = \varphi'_{(x=0)} = \alpha_2$$

$$D_5 = \varphi_{(x=a)} = \alpha_1 + \alpha_2 a + \alpha_3 a^2 + \alpha_4 a^3 + \alpha_5 a^5$$

$$D_6 = \varphi'_{(x=a)} = \alpha_2 + 2\alpha_3 a + 3\alpha_4 a^2 + 5\alpha_5 a^4$$

$$D_7 = \varphi_{(x=\frac{a}{2})} = \alpha_1 + \alpha_2(\frac{a}{2}) + \alpha_3(\frac{a}{2})^2 + \alpha_4(\frac{a}{2})^3 + \alpha_5(\frac{a}{2})^5$$



E.L 3-2

$$D_5 = \varphi_{(x=0)} = \alpha_1$$

$$D_6 = \varphi'_{(x=0)} = \alpha_2$$

$$D_3 = \varphi_{(x=b)} = \alpha_1 + \alpha_2 b + \alpha_3 b^2 + \alpha_4 b^3 + \alpha_5 b^5$$

$$D_4 = \varphi'_{(x=b)} = \alpha_2 + 2\alpha_3 b + 3\alpha_4 b^2 + 5\alpha_5 b^4$$

$$D_8 = \varphi_{(x=\frac{b}{2})} = \alpha_1 + \alpha_2(\frac{b}{2}) + \alpha_3(\frac{b}{2})^2 + \alpha_4(\frac{b}{2})^3 + \alpha_5(\frac{b}{2})^5$$

$$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & a & a^2 & a^3 & a^5 \\ 0 & 1 & 2a & 3a^2 & 5a^4 \\ 1 & \frac{a}{2} & (\frac{a}{2})^2 & (\frac{a}{2})^3 & (\frac{a}{2})^5 \end{bmatrix}$$

$$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & b & b^2 & b^3 & b^5 \\ 0 & 1 & 2b & 3b^2 & 5b^4 \\ 1 & \frac{b}{2} & (\frac{b}{2})^2 & (\frac{b}{2})^3 & (\frac{b}{2})^5 \end{bmatrix}$$

$$[A_E]^{-1} = \begin{array}{c} \begin{array}{ccccc} D_1 & D_2 & D_5 & D_6 & D_7 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -\frac{47}{5a^2} & -\frac{18}{5a} & -\frac{17}{5a^2} & \frac{3}{5a} & \frac{64}{5a^2} \\ \frac{58}{5a^3} & \frac{17}{5a^2} & \frac{38}{5a^3} & -\frac{7}{5a^2} & -\frac{96}{5a^3} \\ -\frac{16}{5a^5} & -\frac{4}{5a^4} & -\frac{16}{5a^5} & \frac{4}{5a^4} & \frac{32}{5a^5} \end{bmatrix} \end{array}$$

$$[A_E]^{-1} = \begin{array}{c} \begin{array}{ccccc} D_5 & D_6 & D_3 & D_4 & D_8 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -\frac{47}{5b^2} & -\frac{18}{5b} & -\frac{17}{5b^2} & \frac{3}{5b} & \frac{64}{5b^2} \\ \frac{58}{5b^3} & \frac{17}{5b^2} & \frac{38}{5b^3} & -\frac{7}{5b^2} & -\frac{96}{5b^3} \\ -\frac{16}{5b^5} & -\frac{4}{5b^4} & -\frac{16}{5b^5} & \frac{4}{5b^4} & \frac{32}{5b^5} \end{bmatrix} \end{array}$$

$$[P] = [1 \quad x \quad x^2 \quad x^3 \quad x^5]$$

$$[N_{(x)}] = [P][A_E]^{-1}$$

$$N''_{1(x)} = -\frac{47}{5a^2}(2) + \frac{58}{5a^3}(6x) - \frac{16}{5a^5}(20x^3)$$
$$N''_{2(x)} = -\frac{18}{5a}(2) + \frac{17}{5a^2}(6x) - \frac{4}{5a^4}(20x^3)$$
$$N''_{3(x)} = -\frac{17}{5a^2}(2) + \frac{38}{5a^3}(6x) - \frac{16}{5a^5}(20x^3)$$
$$N''_{4(x)} = \frac{3}{5a}(2) - \frac{7}{5a^2}(6x) + \frac{4}{5a^4}(20x^3)$$
$$N''_{5(x)} = \frac{64}{5a^2}(2) - \frac{96}{5a^3}(6x) + \frac{32}{5a^5}(20x^3)$$

$$N''_{1(x)} = -\frac{47}{5b^2}(2) + \frac{58}{5b^3}(6x) - \frac{16}{5b^5}(20x^3)$$
$$N''_{2(x)} = -\frac{18}{5b}(2) + \frac{17}{5b^2}(6x) - \frac{4}{5b^4}(20x^3)$$
$$N''_{3(x)} = -\frac{17}{5b^2}(2) + \frac{38}{5b^3}(6x) - \frac{16}{5b^5}(20x^3)$$
$$N''_{4(x)} = \frac{3}{5b}(2) - \frac{7}{5b^2}(6x) + \frac{4}{5b^4}(20x^3)$$
$$N''_{5(x)} = \frac{64}{5b^2}(2) - \frac{96}{5b^3}(6x) + \frac{32}{5b^5}(20x^3)$$

$$[K]_F = EI\int_0^l [N''(x)]^T [N''(x)]dx$$

$$[K]_{1-3} = \begin{array}{c} \begin{array}{ccc} 5 & 6 & 7 \end{array} \\ \begin{bmatrix} \frac{4l(2023a^6 - 13566a^5l + 30324a^4l^2 + 9520a^3l^3 - 51072a^2l^4 + 25600l^6)}{175a^{10}} & \frac{-2l(714a^6 - 4893a^5l + 11172a^4l^2 + 4060a^3l^3 - 22176a^2l^4 + 12800l^6)}{175a^9} & \frac{-64l(476a^6 - 2667a^5l + 4788a^4l^2 + 1715a^3l^3 - 7224a^2l^4 + 3200l^6)}{175a^{10}} \\ \frac{-2l(714a^6 - 4893a^5l + 11172a^4l^2 + 4060a^3l^3 - 22176a^2l^4 + 12800l^6)}{175a^9} & \frac{4l(63a^6 - 441a^5l + 1029a^4l^2 + 420a^3l^3 - 2352a^2l^4 + 1600l^6)}{175a^8} & \frac{64l(84a^6 - 483a^5l + 882a^4l^2 + 385a^3l^3 - 1596a^2l^4 + 800l^6)}{175a^9} \\ \frac{-64l(476a^6 - 2667a^5l + 4788a^4l^2 + 1715a^3l^3 - 7224a^2l^4 + 3200l^6)}{175a^{10}} & \frac{64l(84a^6 - 483a^5l + 882a^4l^2 + 385a^3l^3 - 1596a^2l^4 + 800l^6)}{175a^9} & \frac{4096l(28a^6 - 126a^5l + 189a^4l^2 + 70a^3l^3 - 252a^2l^4 + 100l^6)}{175a^{10}} \end{bmatrix} \end{array} \begin{array}{c} 5 \\ 6 \\ 7 \end{array}$$

$$[K]_{3-2} = \begin{array}{c} \begin{array}{ccc} 5 & 6 & 8 \end{array} \\ \begin{bmatrix} \frac{4l(15463b^6 - 57246b^5l + 70644b^4l^2 + 26320b^3l^3 - 77952b^2l^4 + 25600l^6)}{175b^{10}} & \frac{2l(11844b^6 - 38703b^5l + 41412b^4l^2 + 16660b^3l^3 - 42336b^2l^4 + 12800l^6)}{175b^9} & \frac{-64l(1316b^6 - 5397b^5l + 7308b^4l^2 + 2765b^3l^3 - 8904b^2l^4 + 3200l^6)}{175b^{10}} \\ \frac{2l(11844b^6 - 38703b^5l + 41412b^4l^2 + 16660b^3l^3 - 42336b^2l^4 + 12800l^6)}{175b^9} & \frac{4l(2268b^6 - 6426b^5l + 6069b^4l^2 + 2520b^3l^3 - 5712b^2l^4 + 1600l^6)}{175b^8} & \frac{-128l(252b^6 - 924b^5l + 1071b^4l^2 + 455b^3l^3 - 1218b^2l^4 + 400l^6)}{175b^9} \\ \frac{-64l(1316b^6 - 5397b^5l + 7308b^4l^2 + 2765b^3l^3 - 8904b^2l^4 + 3200l^6)}{175b^{10}} & \frac{-128l(252b^6 - 924b^5l + 1071b^4l^2 + 455b^3l^3 - 1218b^2l^4 + 400l^6)}{175b^9} & \frac{4096l(28b^6 - 126b^5l + 189b^4l^2 + 70b^3l^3 - 252b^2l^4 + 100l^6)}{175b^{10}} \end{bmatrix} \end{array} \begin{array}{c} 5 \\ 6 \\ 8 \end{array}$$

$$[K]_{Structure} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

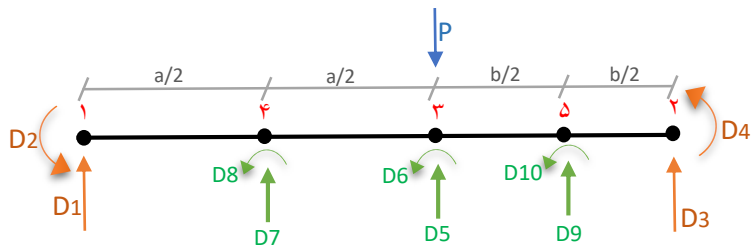
$$[S] =$$

$\frac{4l(2023a^6 - 13566a^5l + 30324a^4l^2 + 9520a^3l^3 - 51072a^2l^4 + 25600l^6)}{175a^{10}} + \frac{4l(15463b^6 - 57246b^5l + 70644b^4l^2 + 26320b^3l^3 - 77952b^2l^4 + 25600l^6)}{175b^{10}}$	$\frac{-2l(714a^6 - 4893a^5l + 11172a^4l^2 + 4060a^3l^3 - 22176a^2l^4 + 12800l^6)}{175a^9} + \frac{2l(11844b^6 - 38703b^5l + 41412b^4l^2 + 16660b^3l^3 - 42336b^2l^4 + 12800l^6)}{175b^9}$	$\frac{-64l(476a^6 - 2667a^5l + 4788a^4l^2 + 1715a^3l^3 - 7224a^2l^4 + 3200l^6)}{175a^{10}}$	$\frac{-64l(1316b^6 - 5397b^5l + 7308b^4l^2 + 2765b^3l^3 - 8904b^2l^4 + 3200l^6)}{175b^{10}}$
$\frac{-2l(714a^6 - 4893a^5l + 11172a^4l^2 + 4060a^3l^3 - 22176a^2l^4 + 12800l^6)}{175a^9} + \frac{2l(11844b^6 - 38703b^5l + 41412b^4l^2 + 16660b^3l^3 - 42336b^2l^4 + 12800l^6)}{175b^9}$	$\frac{4l(63a^6 - 441a^5l + 1029a^4l^2 + 420a^3l^3 - 2352a^2l^4 + 1600l^6)}{175a^8} + \frac{4l(2268b^6 - 6426b^5l + 6069b^4l^2 + 2520b^3l^3 - 5712b^2l^4 + 1600l^6)}{175b^8}$	$\frac{64l(84a^6 - 483a^5l + 882a^4l^2 + 385a^3l^3 - 1596a^2l^4 + 800l^6)}{175a^9}$	$\frac{-128l(252b^6 - 924b^5l + 1071b^4l^2 + 455b^3l^3 - 1218b^2l^4 + 400l^6)}{175b^9}$
$\frac{-64l(476a^6 - 2667a^5l + 4788a^4l^2 + 1715a^3l^3 - 7224a^2l^4 + 3200l^6)}{175a^{10}}$	$\frac{64l(84a^6 - 483a^5l + 882a^4l^2 + 385a^3l^3 - 1596a^2l^4 + 800l^6)}{175a^9}$	$\frac{4096l(28a^6 - 126a^5l + 189a^4l^2 + 70a^3l^3 - 252a^2l^4 + 100l^6)}{175a^{10}}$	0
$\frac{-64l(1316b^6 - 5397b^5l + 7308b^4l^2 + 2765b^3l^3 - 8904b^2l^4 + 3200l^6)}{175b^{10}}$	$\frac{-128l(252b^6 - 924b^5l + 1071b^4l^2 + 455b^3l^3 - 1218b^2l^4 + 400l^6)}{175b^9}$	0	$\frac{4096l(28b^6 - 126b^5l + 189b^4l^2 + 70b^3l^3 - 252b^2l^4 + 100l^6)}{175b^{10}}$

$$D_5: P_{D_5} = -P$$
$$D_6: P_{D_6} = 0$$
$$D_7: P_{D_7} = 0$$
$$D_8: P_{D_8} = 0$$

$$[S]\{D\} = \{P\} \equiv [K]\{U\} = \{P\}$$

$$[S]\begin{Bmatrix} D_5 \\ D_6 \\ D_7 \\ D_8 \end{Bmatrix} = \begin{Bmatrix} -P \\ 0 \\ 0 \\ 0 \end{Bmatrix} \rightarrow D_5, D_6, D_7, D_8 \ \checkmark\checkmark \ , \quad P_x = (-P)N_{(x)} \rightarrow P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8 \ \checkmark\checkmark$$



Mesh of
4 lower order E.B Beam F.E

E.L 1-4

$$[K]_{1-4} = \begin{array}{c} \begin{array}{cccc} D_1 & D_2 & D_7 & D_9 \end{array} \\ \begin{bmatrix} \frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} & -\frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} \\ \frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} \\ -\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} \\ \frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})} \end{bmatrix} \end{array}$$

E.L 4-3

$$[K]_{4-3} = \begin{array}{c} \begin{array}{cccc} D_7 & D_9 & D_5 & D_6 \end{array} \\ \begin{bmatrix} \frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} & -\frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} \\ \frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} \\ -\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} \\ \frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})} \end{bmatrix} \end{array}$$

E.L 3-5

$$[K]_{3-5} = \begin{bmatrix} \frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} & -\frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} \\ \frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} \\ -\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} \\ \frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})} \end{bmatrix}$$

E.L 5-2

$$[K]_{5-2} = \begin{bmatrix} \frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} & -\frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} \\ \frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} \\ -\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} \\ \frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})} \end{bmatrix}$$

$$[K]_{1-3} = \begin{array}{c} \begin{array}{cccccc} D1 & D2 & D7 & D9 & D5 & D6 \end{array} \\ \begin{bmatrix} \frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} & -\frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} & 0 & 0 \\ \frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & 0 & 0 \\ -\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{12EI}{(\frac{a}{2})^3}+\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{\left(\frac{a}{2}\right)^2}+\frac{6EI}{(\frac{a}{2})^2} & -\frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} \\ \frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2}+\frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})}+\frac{4EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} \\ 0 & 0 & -\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} \\ 0 & 0 & \frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})} \end{bmatrix} \end{array}$$

$$[K]_{3-5} = \begin{array}{c} \begin{array}{cccccc} D5 & D6 & D8 & D10 & D3 & D4 \end{array} \\ \begin{bmatrix} \frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} & -\frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} & 0 & 0 \\ \frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & 0 & 0 \\ -\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{12EI}{(\frac{b}{2})^3}+\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{\left(\frac{b}{2}\right)^2}+\frac{6EI}{(\frac{b}{2})^2} & -\frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} \\ \frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2}+\frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})}+\frac{4EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} \\ 0 & 0 & -\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} \\ 0 & 0 & \frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})} \end{bmatrix} \end{array}$$

$$
[K]_{structure} = \begin{array}{c}
\begin{array}{cccccccccc} D1 & D2 & D7 & D9 & D5 & D6 & D8 & D10 & D3 & D4 \end{array} \\
\begin{bmatrix}
\frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} & -\frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{12EI}{(\frac{a}{2})^3}+\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{\left(\frac{a}{2}\right)^2}+\frac{6EI}{(\frac{a}{2})^2} & -\frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} & 0 & 0 & 0 & 0 \\
\frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2}+\frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})}+\frac{4EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{12EI}{(\frac{a}{2})^3}+\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{\left(\frac{a}{2}\right)^2}+\frac{6EI}{(\frac{b}{2})^2} & -\frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} & 0 & 0 \\
0 & 0 & \frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2}+\frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{a}{2})}+\frac{4EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{12EI}{(\frac{b}{2})^3}+\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{\left(\frac{b}{2}\right)^2}+\frac{6EI}{(\frac{b}{2})^2} & -\frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} \\
0 & 0 & 0 & 0 & \frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2}+\frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})}+\frac{4EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})}
\end{bmatrix}
\end{array}
\begin{array}{c} \\ D1 \\ D2 \\ D7 \\ D9 \\ D5 \\ D6 \\ D8 \\ D10 \\ D3 \\ D4 \end{array}
$$

$$D_5: P_{D_5} = -P$$
$$D_6: P_{D_6} = 0$$
$$D_7: P_{D_7} = 0$$
$$D_8: P_{D_8} = 0$$
$$D_9: P_{D_9} = 0$$
$$D_{10}: P_{D_{10}} = 0$$

$$[S]\{D\} = \{P\} \equiv [K]\{U\} = \{P\}$$

$$\begin{array}{c}
\begin{array}{cccccc} D7 & D9 & D5 & D6 & D8 & D10 \end{array} \\
\begin{bmatrix}
2*\frac{12EI}{\left(\frac{a}{2}\right)^3} & 0 & -\frac{12EI}{\left(\frac{a}{2}\right)^3} & \frac{6EI}{\left(\frac{a}{2}\right)^2} & 0 & 0 \\
0 & 2*\frac{4EI}{\left(\frac{a}{2}\right)} & -\frac{6EI}{\left(\frac{a}{2}\right)^2} & \frac{2EI}{\left(\frac{a}{2}\right)} & 0 & 0 \\
-\frac{12EI}{\left(\frac{a}{2}\right)^3} & -\frac{6EI}{\left(\frac{a}{2}\right)^2} & 2*\frac{12EI}{\left(\frac{a}{2}\right)^3} & -\frac{6EI}{\left(\frac{a}{2}\right)^2}+\frac{6EI}{\left(\frac{b}{2}\right)^2} & -\frac{12EI}{\left(\frac{b}{2}\right)^3} & \frac{6EI}{\left(\frac{b}{2}\right)^2} \\
\frac{6EI}{\left(\frac{a}{2}\right)^2} & \frac{2EI}{\left(\frac{a}{2}\right)} & -\frac{6EI}{\left(\frac{a}{2}\right)^2}+\frac{6EI}{\left(\frac{b}{2}\right)^2} & \frac{4EI}{\left(\frac{a}{2}\right)}+\frac{4EI}{\left(\frac{b}{2}\right)} & -\frac{6EI}{\left(\frac{b}{2}\right)^2} & \frac{2EI}{\left(\frac{b}{2}\right)} \\
0 & 0 & -\frac{12EI}{\left(\frac{b}{2}\right)^3} & -\frac{6EI}{\left(\frac{b}{2}\right)^2} & 2*\frac{12EI}{\left(\frac{b}{2}\right)^3} & 0 \\
0 & 0 & \frac{6EI}{\left(\frac{b}{2}\right)^2} & \frac{2EI}{\left(\frac{b}{2}\right)} & 0 & 2*\frac{4EI}{\left(\frac{b}{2}\right)}
\end{bmatrix}
\end{array}
\begin{Bmatrix} D_7 \\ D_9 \\ D_5 \\ D_6 \\ D_8 \\ D_{10} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ -P \\ 0 \\ 0 \\ 0 \end{Bmatrix} \rightarrow D_5, D_6, D_7, D_8, D_9, D_{10} \ \surd\surd$$

$$\{\boldsymbol{P_x}\} = [\boldsymbol{K}]_{\boldsymbol{E.l}}\{\boldsymbol{D_x}\}$$

$$\begin{Bmatrix} \boldsymbol{P_1} \\ \boldsymbol{P_2} \\ \boldsymbol{P_7} \\ \boldsymbol{P_9} \\ \boldsymbol{P_5} \\ \boldsymbol{P_6} \end{Bmatrix} = \begin{bmatrix} \overset{D1}{\frac{12EI}{(\frac{a}{2})^3}} & \overset{D2}{\frac{6EI}{(\frac{a}{2})^2}} & \overset{D7}{-\frac{12EI}{(\frac{a}{2})^3}} & \overset{D9}{\frac{6EI}{(\frac{a}{2})^2}} & \overset{D5}{0} & \overset{D6}{0} \\ \frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & 0 & 0 \\ -\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{12EI}{(\frac{a}{2})^3}+\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{\left(\frac{a}{2}\right)^2}+\frac{6EI}{(\frac{a}{2})^2} & -\frac{12EI}{(\frac{a}{2})^3} & \frac{6EI}{(\frac{a}{2})^2} \\ \frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2}+\frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})}+\frac{4EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} \\ 0 & 0 & -\frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{12EI}{(\frac{a}{2})^3} & -\frac{6EI}{(\frac{a}{2})^2} \\ 0 & 0 & \frac{6EI}{(\frac{a}{2})^2} & \frac{2EI}{(\frac{a}{2})} & -\frac{6EI}{(\frac{a}{2})^2} & \frac{4EI}{(\frac{a}{2})} \end{bmatrix} \begin{Bmatrix} D_1 = 0 \\ D_2 = 0 \\ D_7 \\ D_9 \\ D_5 \\ D_6 \end{Bmatrix} \rightarrow P_1, P_2, P_5, P_6, P_7, P_9 \ \surd\surd$$

$$\begin{Bmatrix} \boldsymbol{P_5} \\ \boldsymbol{P_6} \\ \boldsymbol{P_8} \\ \boldsymbol{P_{10}} \\ \boldsymbol{P_3} \\ \boldsymbol{P_4} \end{Bmatrix} = \begin{bmatrix} \overset{D5}{\frac{12EI}{(\frac{b}{2})^3}} & \overset{D6}{\frac{6EI}{(\frac{b}{2})^2}} & \overset{D8}{-\frac{12EI}{(\frac{b}{2})^3}} & \overset{D10}{\frac{6EI}{(\frac{b}{2})^2}} & \overset{D3}{0} & \overset{D4}{0} \\ \frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & 0 & 0 \\ -\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{12EI}{(\frac{b}{2})^3}+\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{\left(\frac{b}{2}\right)^2}+\frac{6EI}{(\frac{b}{2})^2} & -\frac{12EI}{(\frac{b}{2})^3} & \frac{6EI}{(\frac{b}{2})^2} \\ \frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2}+\frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})}+\frac{4EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} \\ 0 & 0 & -\frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{12EI}{(\frac{b}{2})^3} & -\frac{6EI}{(\frac{b}{2})^2} \\ 0 & 0 & \frac{6EI}{(\frac{b}{2})^2} & \frac{2EI}{(\frac{b}{2})} & -\frac{6EI}{(\frac{b}{2})^2} & \frac{4EI}{(\frac{b}{2})} \end{bmatrix} \begin{Bmatrix} D_5 \\ D_6 \\ D_8 \\ D_{10} \\ D_3 = 0 \\ D_4 = 0 \end{Bmatrix} \rightarrow P_3, P_4, P_5, P_6, P_8, P_{10} \ \surd\surd$$